

Priya Abiram: Fluids Blade Design Project Overview

Project Overview

This project focused on the design, fabrication, and experimental testing of a 6-inch horizontal-axis wind turbine blade for laboratory wind-tunnel operation. The goal was to maximize aerodynamic power extraction while satisfying geometric, structural, and manufacturability constraints, including a 2-inch axial projection limit, a 44 MPa flexural stress limit, and 3D-printing feasibility. The blade was designed to operate efficiently within wind speeds generated by the course fan system.

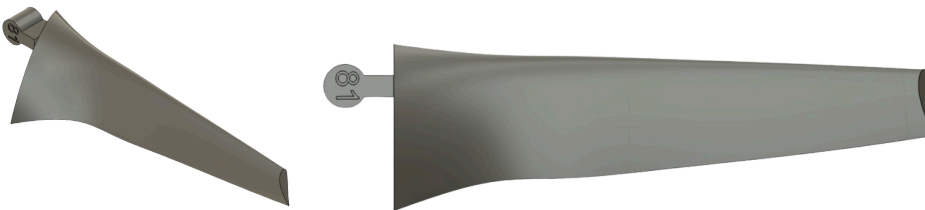
Design Process

An iterative Blade Element Momentum Theory (BEMT) approach was used to develop the blade geometry. NACA 4412 airfoil data were imported at multiple Reynolds numbers and interpolated to match local flow conditions. An energy-weighted design wind speed of 5.04 m/s was selected using a Weibull distribution. Initial chord and twist distributions were refined through full-span BEMT convergence across 30 blade stations. Numerical instabilities near the root and tip were addressed using quadratic and linear smoothing, respectively, producing a continuous and manufacturable geometry. Structural stresses were verified using flapwise bending estimates and remained well below allowable limits.

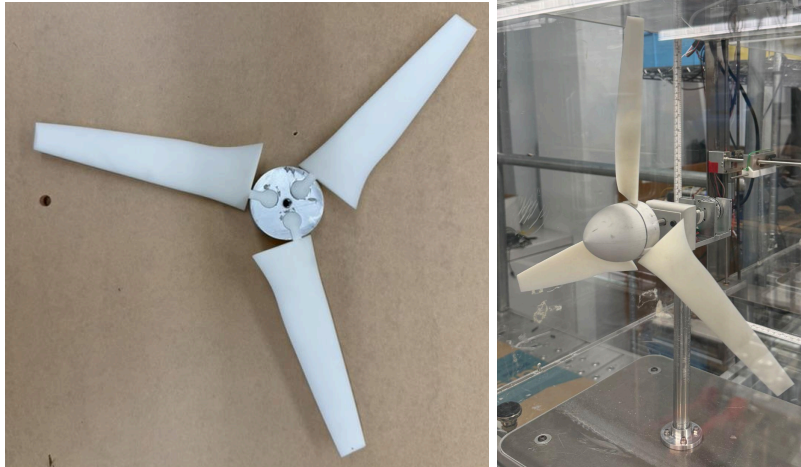
Testing Summary

The blade was 3D-printed, mounted in a hub, and tested in a wind tunnel using a torque-brake system. Fan frequency, wind speed, rotational speed, and brake current were recorded during steady-state sweep-up tests. Power, torque, tip-speed ratio (TSR), and power coefficient were computed and compared against design predictions. The blade achieved a maximum measured power of 1.29 W at 5.56 m/s and consistently operated near $TSR \approx 6$, exceeding the original design TSR of 3. Resonance effects were observed between approximately 2100–2500 RPM but did not compromise structural integrity.

Blades outside hub:

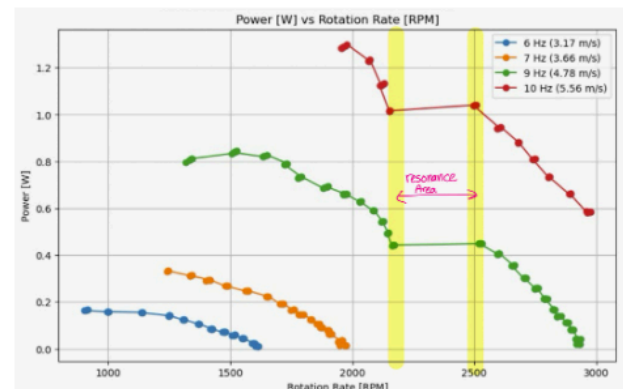
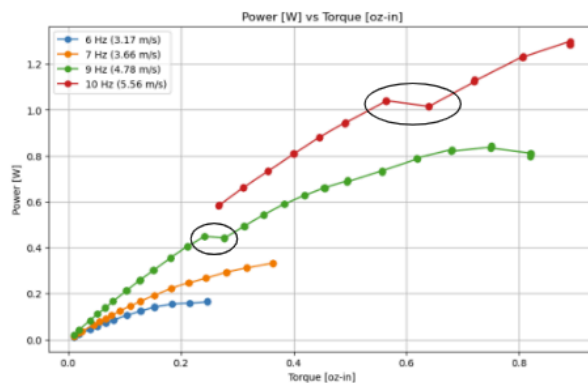


Blades in Hub:



Experimental Power Output

- Maximum measured power: 1.29 watts at 5.56 m/s
- Power increased consistently with wind speed, without showing a clear peak within the tested range
- A resonance region occurred between about 2100 and 2500 RPM, temporarily lowering measured power



Max Power = 1.29W at 5.56m/s

Although the blade was designed for $TSR = 3$, the experimental turbine consistently operated near $TSR = 6$. Despite this mismatch, the performance remained strong. This suggests the blade could be redesigned for a higher design TSR in future iterations. Predicted torque was $0.0117 \text{ N}\cdot\text{m}$, with a predicted power of approximately 1 watt at 812 RPM. The experimental blade exceeded this prediction because it operated at significantly higher RPM and TSR .

The blade exhibited no structural failure or undesirable deformation during testing.

My Contribution

I led the aerodynamic modeling and BEMT implementation, Reynolds number interpolation, and root/tip smoothing. I also performed structural stress verification, contributed to performance prediction, and supported experimental data analysis and interpretation.